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# **Software Requirements Specification**

**for**

**SajiloDocs**

**Version 1.0 approved**

**Prepared by**

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# **1 Introduction**

## **1.1 Purpose**

This document outlines the software requirements for SajiloDocs. It serves as a comprehensive guide for building a secure digital locker system. The document highlights the key features, technical requirements, and performance expectations needed to create a platform where users can safely store, manage, and translate their personal documents.

## **1.2 Document Conventions**

This specification follows the standard IEEE formatting guidelines for software requirements. System features are ranked by priority (High, Medium, Low) to help the development team understand which core functions need to be built first.

## **1.3 Intended Audience and Reading Suggestions**

This document is written for the project's development team, project managers, and the evaluating instructors at Itahari International College. Readers should start with the overall description to understand the project's big picture before moving on to the specific technical features.

## **1.4 Product Scope**

SajiloDocs is a web application created to solve the common problem of misplacing important physical documents like academic records, property papers, and identification cards. While general cloud storage exists, this platform is specifically tailored for Nepali users. It offers localized features like extracting text from Nepali documents using OCR, automatically translating them into English, and sending reminders when a document is about to expire. The main goal is to reduce manual work and keep sensitive information highly secure and organized.

## **1.5 References**

ISO/IEC/IEEE, 2018. ISO/IEC/IEEE 29148:2018 - Systems and software engineering - Life cycle processes - Requirements engineering. Geneva, Switzerland: International Organization for Standardization.

## **2 Overall Description**

### **2.1 Product Perspective**

SajiloDocs will be developed as a brand-new, standalone web platform. The system architecture uses React for the user interface, Django for managing backend operations, and MySQL for organizing the database.

### **2.2 Product Functions**

The application will perform several major functions:

- Registering and authenticating users safely.
- Allowing users to upload, store, and categorize their files with custom tags.
- Sharing documents with specific permission levels, such as view, download, or edit.
- Processing scanned documents to extract text and detect key fields.
- Translating text into English and re-creating it as a formatted PDF.
- Sending automated notifications before a document expires.

### **2.3 User Classes and Characteristics**

The target users are everyday individuals, students, professionals, and small business owners. Because the system manages important files, users simply need a basic understanding of how to use a web browser and upload files from their computer or mobile device.

### **2.4 Operating Environment**

SajiloDocs is a web-based service, meaning it will run smoothly on standard web browsers like Chrome, Safari, and Firefox. The server side requires an environment that supports Python, Django, and a MySQL database system.

### **2.5 Design and Implementation Constraints**

The project must strictly stick to the chosen technology stack (React, Django, MySQL). Additionally, we are required to use specific libraries for certain tasks, such as Tesseract for text extraction and either the Google Cloud or DeepL API for translations.

## 2.6 User Documentation

Once finished, the project will include a detailed user manual and API documentation to help users and future developers understand how the system works.

## 2.7 Assumptions and Dependencies

SajiloDocs have been established and are operating under the following assumptions and dependencies:

- There is a constant internet connection which the users can utilize to access the web application.
- Users have the simplest knowledge about the use of the web browsers and file uploading.
- It relies on third-party software like Tesseract OCR to extract the text and translators like Google Cloud or DeepL to translate the language into other ones.
- The quality of the uploaded scanned documents determines the accuracy of the results of OCR.
- There should be email notification services where expiry alerts are sent.
- The system presupposes that the user inserts appropriate metadata like issue date and expiry date.
- The platform relies on the well-configured server behind Django, React, and MySQL.

Such assumptions guarantee full functionality but can have an impact on the functionality of systems in the case not met.

### **3 External Interface Requirements**

#### **3.1 User Interfaces**

The frontend will be built with React and styled using Tailwind CSS to ensure it looks clean and works well on different screen sizes. Key screens include a login portal, a personal dashboard for managing files, and interactive interfaces for reviewing translations and OCR results.

#### **3.2 Hardware Interfaces**

SajiloDocs is a web-based solution and it does not need a specific hardware. It works with all types of standard computing gadgets including desktops, laptops, tablets, and smartphones. Users are able to post documents in the local storage or take images with the device cameras. The system is connected to hardware devices including input/output peripherals and storage devices via standard browser interfaces. There is no hardware level integration needed and all the communication is done via the web browser and operating system.

#### **3.3 Software Interfaces**

Database Connection: The backend will communicate with the MySQL database using Django's built-in tools.

#### **3.4 Communications Interfaces**

The system will connect to Python Tesseract for text recognition and third-party translation APIs to handle language conversion.

## 4 System Features

This are the major services to be provided by the system after the development.

### 4.1 System Feature 1 : Account Management and Security

#### 4.1.1 Description and Priority

High Priority. This handles how users access the system securely.

#### 4.1.2 Stimulus/Response Sequences

- Stimulus: User provides the registration information and clicks on a form.  
Response: Checks System input, creates an account and redirects to dashboard.
- Stimulus: Clicking on the credentials of the log-in.  
Response: System verification and admits access with valid credentials show an error with no valid credentials.
- Stimulus: Tries to use dash board but is not an authorized user.  
Response: SYS was redirected to a log in page.

#### 4.1.3 Functional Requirements

REQ-1: The software must allow users to sign up and log in using an email and password, or through Google sign-in.

REQ-2: The system needs to provide a private dashboard showing the user's uploaded and shared files.

### 4.2 System Feature 2 : Document Handling

#### 4.2.1 Description and Priority

High Priority. This is the main functionality for storing and tracking files.

#### 4.2.2 Stimulus/Response Sequences

- Stimulus: clicking on document file.  
response: register and store you in database.
- Stimulus: metadata of a user (issue date and expiry date).  
Response: Metadata of the document stored is in the system.
- Stimulus: Clicking a document folder by the user.

Response: Systemic alteration of the system.

#### **4.2.3 Functional Requirements**

REQ-1: Users must be able to upload files in common formats like PDF, JPG, PNG, and DOCX.

REQ-2: The system should let users organize these files into specific folders.

REQ-3: Users must be able to add helpful details to their files, such as the issue date and expiry date.

### **4.3 System Feature 3 : Text Extraction and Translation**

#### **4.3.1 Description and Priority**

High Priority. allows the system to intelligently read and translate Nepali documents.

#### **4.3.2 Stimulus/Response Sequences**

- Stimulus: Stimulus extraction text, is taken in an interesting manner.  
Response: The number of the documents, dates among others is identified by the system.
- Stimulus: Localization of the input of the user.  
Response: Text preview and translation of texts.
- Stimulus: the user becomes aware of the fact that it is translated.  
Response: PDF is a system that is automatically generated.

#### **4.3.3 Functional Requirements**

REQ-1: The platform will use OCR technology to read text from uploaded images.

REQ-2: It must automatically recognize important details like document numbers and dates.

REQ-3: The system must translate the read text into English and give users a chance to review it.

REQ-4: After reviewing, users should be able to download the final translation as a well-formatted PDF.

### **4.4 System Feature 4 : File Sharing Controls**

#### **4.4.1 Description and Priority**

Medium Priority. This handles how users safely share information.



#### 4.4.2 Stimulus/Response Sequences

- Stimulus: Sharing link is a user generated one.  
Response: There is responsibility in the aspect of gaining access and secure network connection.
- Stimulus: status is accessed by another user.  
Response: Access permission of the system.
- Stimulus: Too much cancellation: Stimulus.  
Response: The common links are the associated system block links that are instant.

#### 4.4.3 Functional Requirements

REQ-1: Users should be able to generate a secure link to share specific documents..

REQ-2: The owner must be able to limit what others can do, such as setting the file to view-only or allowing edits.

REQ-3: Users must have the ability to cancel these sharing permissions at any time.

### 4.5 System Feature 5 : Expiry Alerts

#### 4.5.1 Description and Priority

Medium Priority. This keeps users informed about renewals.

#### 4.5.2 Stimulus/Response Sequences

- Stimulus: Doc storage- Expiry date.  
Response: Storage of automatic expiry dates in the system.
- Stimulus: Expiry date is near.  
Response: This is notified (via email or in-app notification) by system.
- Stimulus: Turned off Notification to the user.  
Response: The system does not have any alerts on the same document anymore.

#### 4.5.3 Functional Requirements

REQ-1: The system needs to automatically track document expiry dates.

REQ-2: Users should receive alerts via email or inside the app when a deadline is approaching.

REQ-3: Users must have the ability to cancel these sharing permissions at any time.

## **5 Other Nonfunctional Requirements**

### **5.1 Performance Requirements**

Tasks like extracting text and translating documents require processing power. The system should handle these requests efficiently so the user does not experience long loading times. Searching through documents using text should also return results very quickly.

### **5.2 Safety Requirements**

It will ensure the data is not lost, they will not be accessed without permission and the misuse of sensitive documents through the system.

To ensure safety:

Data backup will also have systems installed to ensure that the data is not lost accidentally.

To avoid corruption or malicious files, files uploads will be checked by the system.

The use of secure session management will be avoided so as to avoid illegal access.

The account holders will also be equipped with power to reclaim their accounts in case of safe authentication.

These solutions will provide some degree of protection in the sense that in case of system failure, it will result loss of user data through accidental or ill intentional causes.

### **5.3 Security Requirements**

Because users are uploading sensitive legal and personal forms, keeping data safe is the top priority. All files and database connections must be heavily encrypted. The system also needs to keep track of who views or edits a document to prevent unauthorized access.

### **5.4 Software Quality Attributes**

The platform must be highly reliable. If an expiry date is entered, the notification system must trigger without failure. It also needs to be incredibly user-friendly so that people who are not good with technology can easily navigate their files.

### **5.5 Business Rules**

The system can be taken to be the following business rules:

- The registered users are only the ones who can upload the document, manage or share.
- The owner is the owner of what he/she posted and he/she has the absolute control over the same.
- Only the authorized users have permission to view, download or edit such actions.
- The owner of document may revoke document sharing.

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- The notifications are automatically activated by the expiry date stored.
- The only document formats that undergo OCR and translation processes are supported ones.
- The user has to use a valid metadata (e.g. dates) to get the system properly functioning.
- These rules provide access control, integrity of the data and reasonable system behaviour.

## 6 Other Requirements

### Database Requirements

The data that was organized in the system ought to have consisted of:

- User information
- Document metadata
- OCR extracted text
- Translation data
- Trading permission and access control.
- Notification records

The database should be consistent, it should have an integrity and easy to access data.

### Internationalization Requirements

The system must support:

- OCR input (nepali language)
- Production English language (printers, translation, UI)

This will provide usability to the local users and long term access.

### Legal Requirements

The system must comply with:

- Principles of data privacy, data protection.
- Handling and storage of legal and personal records in a safe manner.
- Restricted unauthorised access of user information.

### Scalability and Reusability Requirement.

The system is supposed to be designed in module fashion where:

- It can be enhanced with new options (e.g. mobile application, offline usage).
- Re-use of APIs and integration with other platforms is possible.

## 7 Appendix A: Glossary

| Term               | Definition                                                                                  |
|--------------------|---------------------------------------------------------------------------------------------|
| OCR                | Optical Character Recognition is the method of an image text retrieval.                     |
| Metadata           | Saved information regarding a document (dates, and description).                            |
| RBAC               | Role-Based Access Control on permissions.                                                   |
| API                | System communication Application programming interface.                                     |
| Dashboard          | This is the area of the display of the features and the documents alongside the user input. |
| Document Vault     | Storage system which is secured for managing documents                                      |
| Translation Engine | This is a translator which may be employed to translate Nepali to English                   |

## 8 Appendix B: Analysis Models

The system comprehended and architecture designed models of analysis are:

- Use Case Diagram: This diagram depicts the relationship between the users and the system.
- Entity Relationship Diagram (ERD): It is the database format and association.
- System architecture diagram: This is a front, back and database communication.
- Flowcharts: document about the flow of the essential functionality components like the uploading of documents and OCR processing.

The models can be used to visualize the system design to maximise on the knowledge of system functioning.

## 9 Appendix C: To Be Determined List

The list of the items that will be finalized or will be enhanced in the future is the following:

- Option of translation API (Google Cloud or DeepL) final.
- Individual enhancement in OCR processing.
- Maximization of intervals of notification.
- Other security systems like multi-factor authentication.
- Offline Support Mobile application.

These things will be revised and once more in the next stages of development.